

Figure 1: Standard connectors with added pins

the xHCI driver. xHCI is the new Host Controller specification developed to support the USB 3.0 transport. The new driver layers look like the diagram in Figure 3.

The code patch is yet to get integrated with the main-line kernel. This is because the xHCI specification has not yet been made public. You can read more on the Linux Super Speed demo in the references mentioned below.

What can end users expect?

The real question now is: as a user of this technology, what can you expect? Most of us use thumb drives and mobile gadgets on a daily basis. We use these to copy files to and from our PCs. These copy processes take more time. With USB 3.0, in theory, the copy process could get 10 times faster. This means that with USB 3.0, you could copy a file that took 30 minutes, within 3 minutes. It has been demonstrated that you get transfer speeds that are at least three times faster, including latencies (hardware/software).

To put it in more technical terms, in the SuperSpeed Conference held last November in San Jose, California, the promoters have demonstrated transfer speeds of 318 MBps on Windows and 233 MBps (typically 125 MBps) on Linux. Compared to this, the file transfer speed of the current USB 2.0 is around 60 MBps. So as an end user, you should buckle up for better speeds while using these gadgets, which are expected to hit the market in 2010. PCs that support these gadgets could also hit the market at the same time.

New technologies introduce more changes in our life. New trends like cheaper memory enabled us to see the entry of gadgets like thumb drives and high-capacity media players. High-speed Internet connections enabled many of us to experience new features like voice over Internet protocol and podcasts.

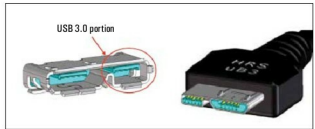


Figure 2: Connectors for hand-held devices

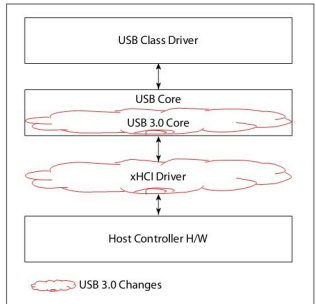


Figure 3: The new driver's layer in USB 3.0

As a popular peripheral connectivity solution, we hope higher speed USB 3.0 brings about better user experiences, apart from other benefits. 

References

- SuperSpeed USB Developers Conference Presentations: www.usb.org/developers/presentations
- USB 3.0 and Linux: sarah.thesharps.us/2008-12-07-13-35.cherry

By: Rajaram Regupathy

The author welcomes your comments and feedback at rera_raja@yahoo.com

Got A LUG?

Visit www.openitIs.com

